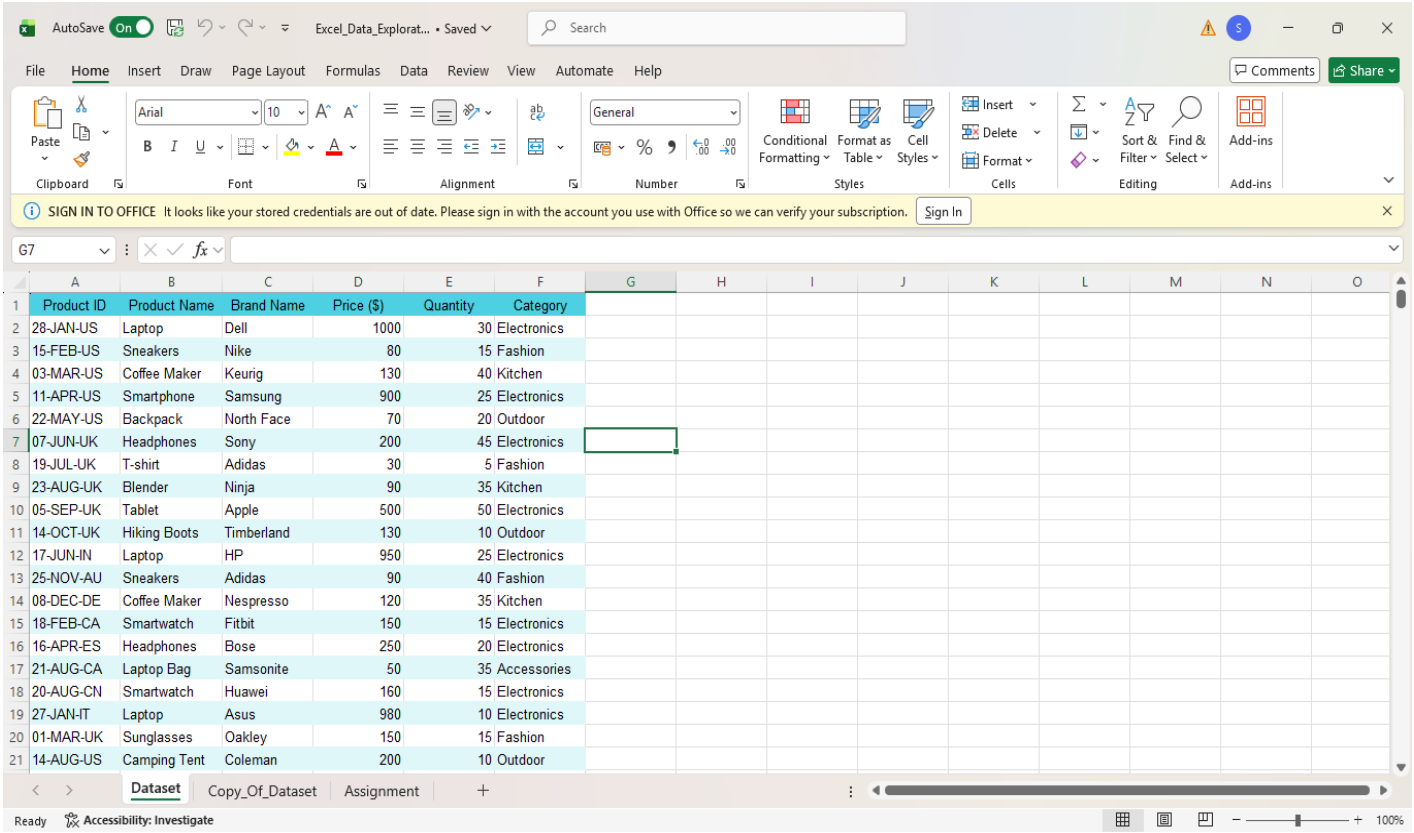
Excel Assignment 1 – Data Exploration

Tasks:

Basic Data Exploration

Converting raw dataset into table and given Table Name as ***"Sales_DB"***.



Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor

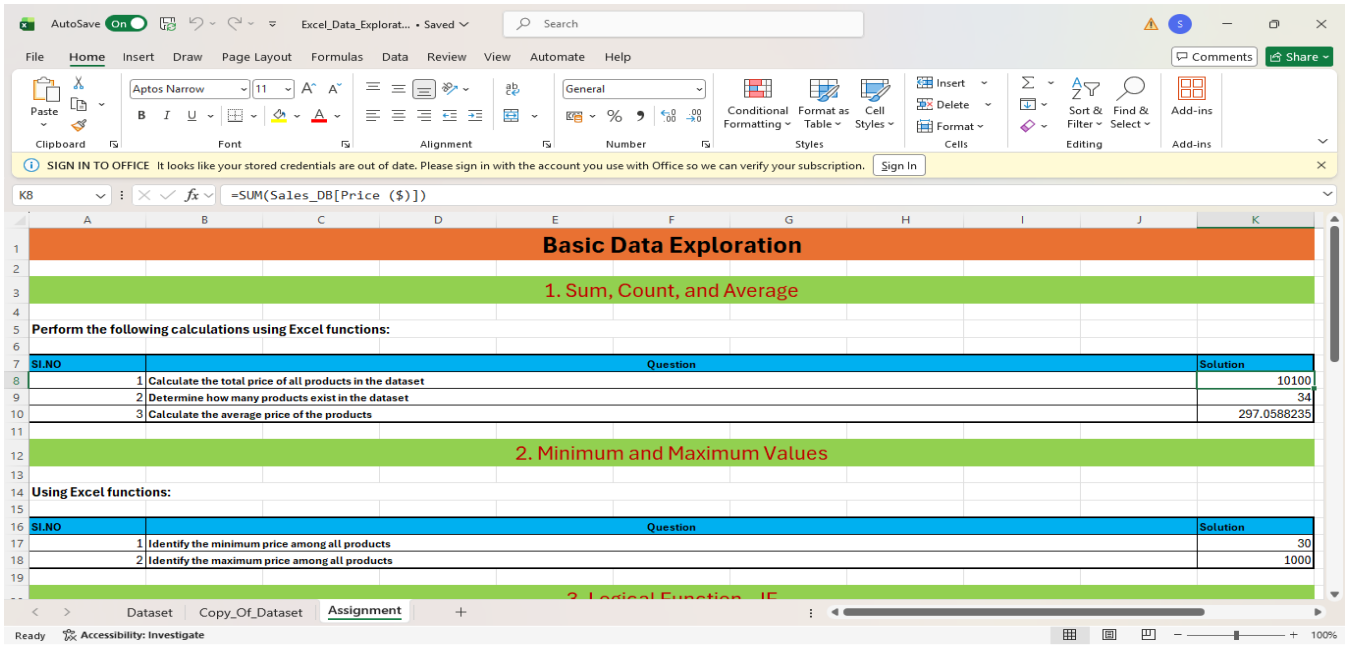
1. Sum, Count, and Average

Data Analytics (DA)

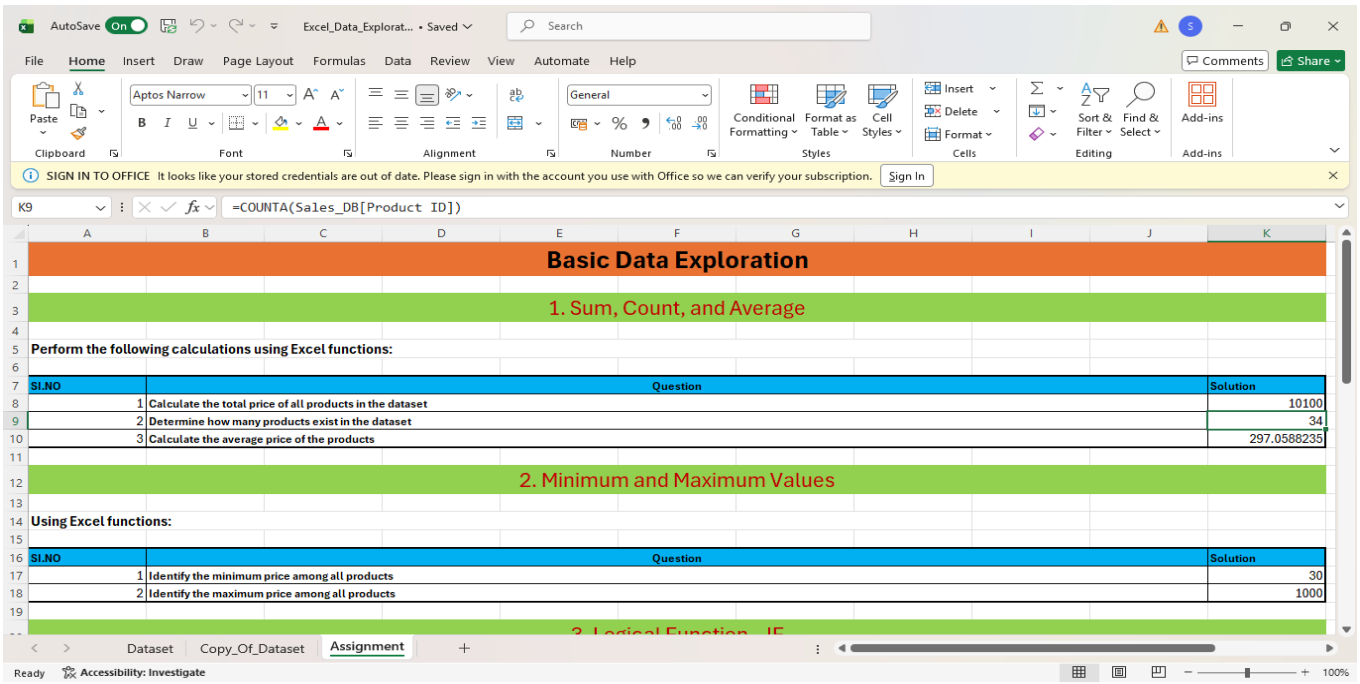
Excel Assignment 1 – Data Exploration

Perform the following calculations using Excel functions:

- Calculate the **total price of all products** in the dataset.

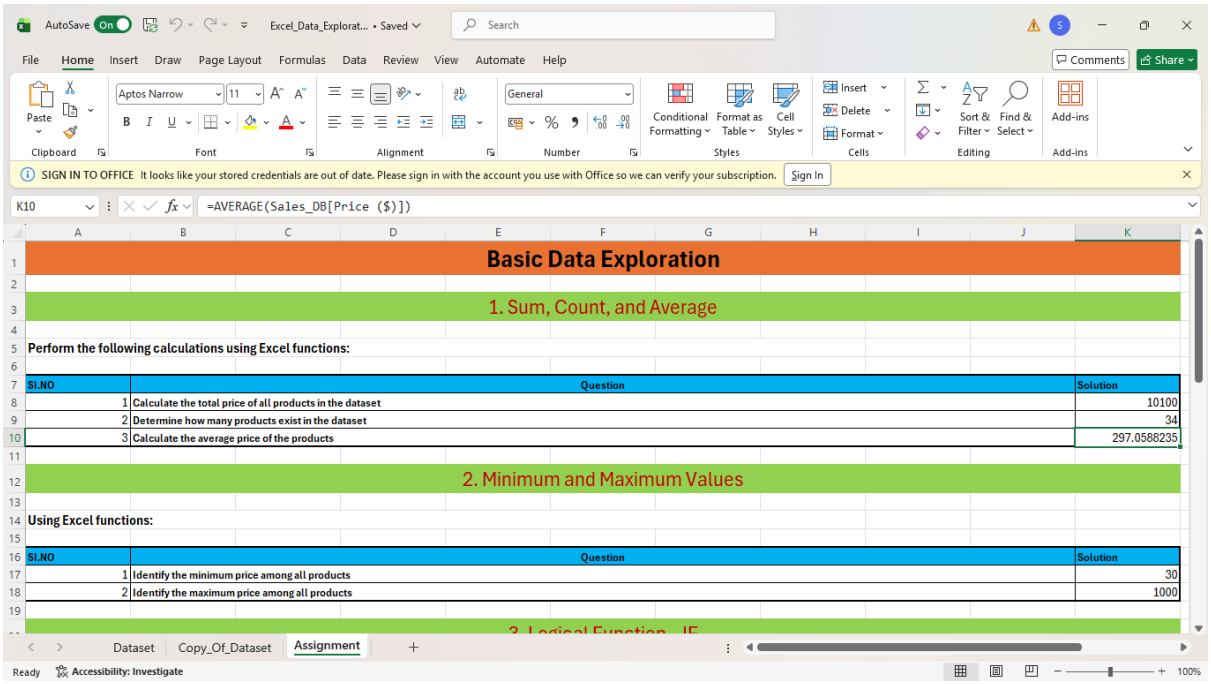


- Determine **how many products exist in the dataset.**



Excel Assignment 1 – Data Exploration

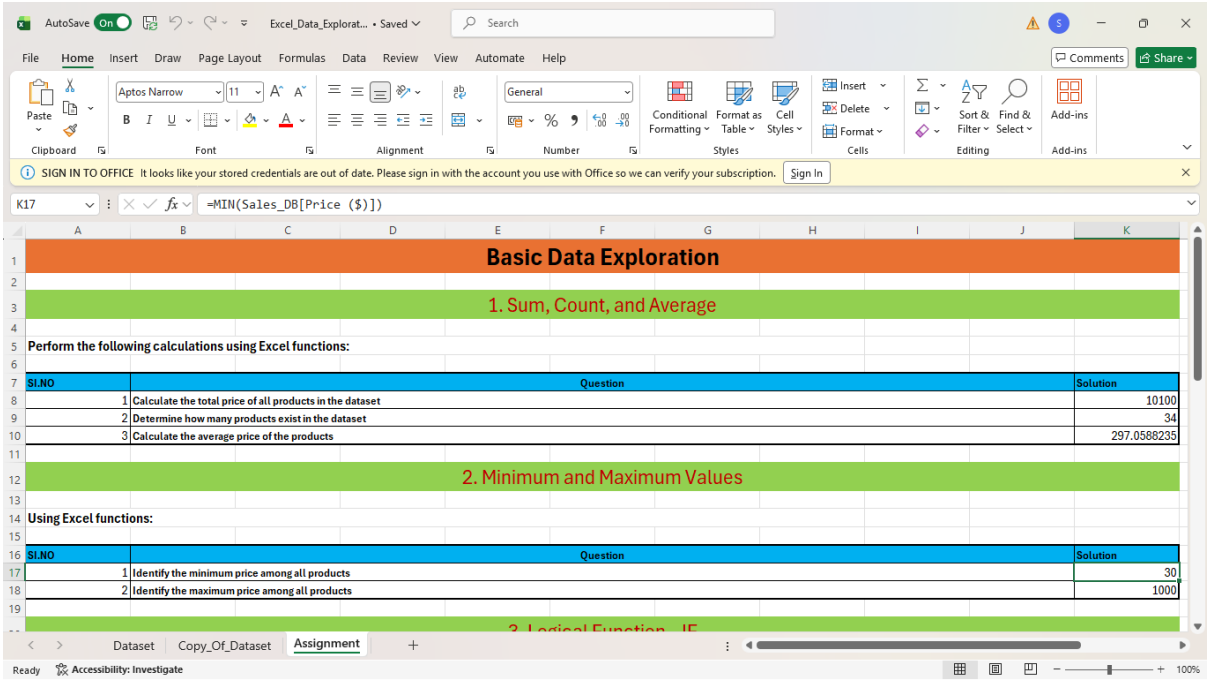
- Calculate the **average price of the products**.



2. Minimum and Maximum Values

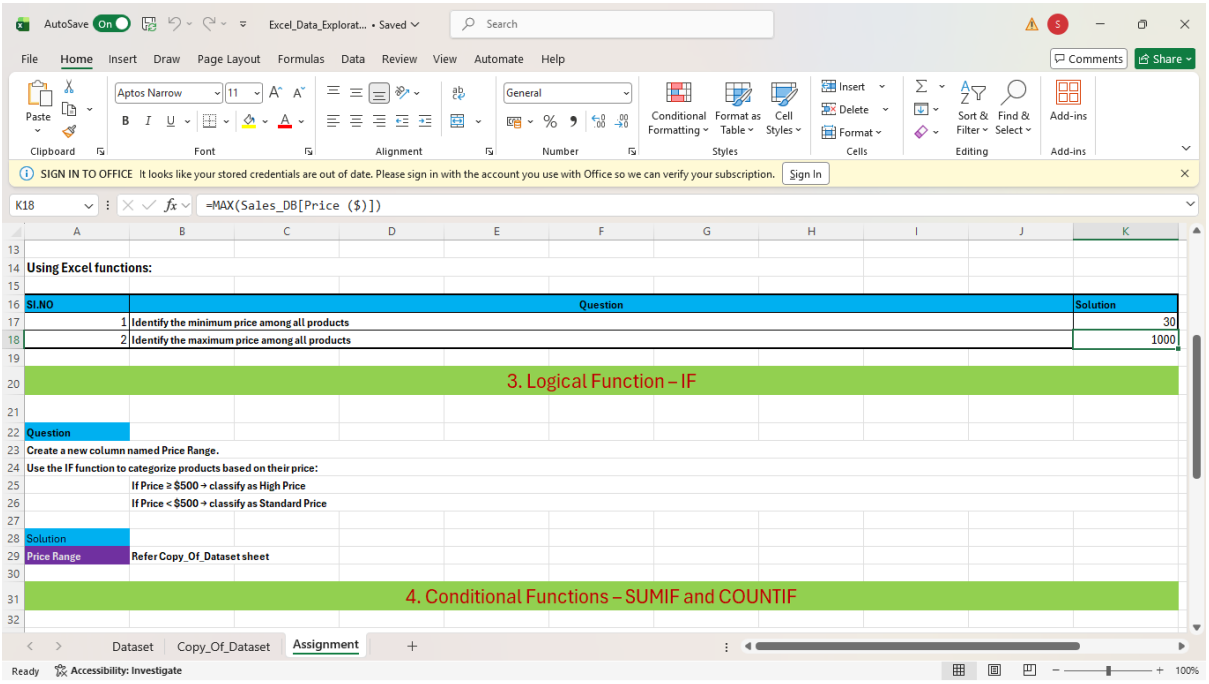
Using Excel functions:

- Identify the **minimum price** among all products.



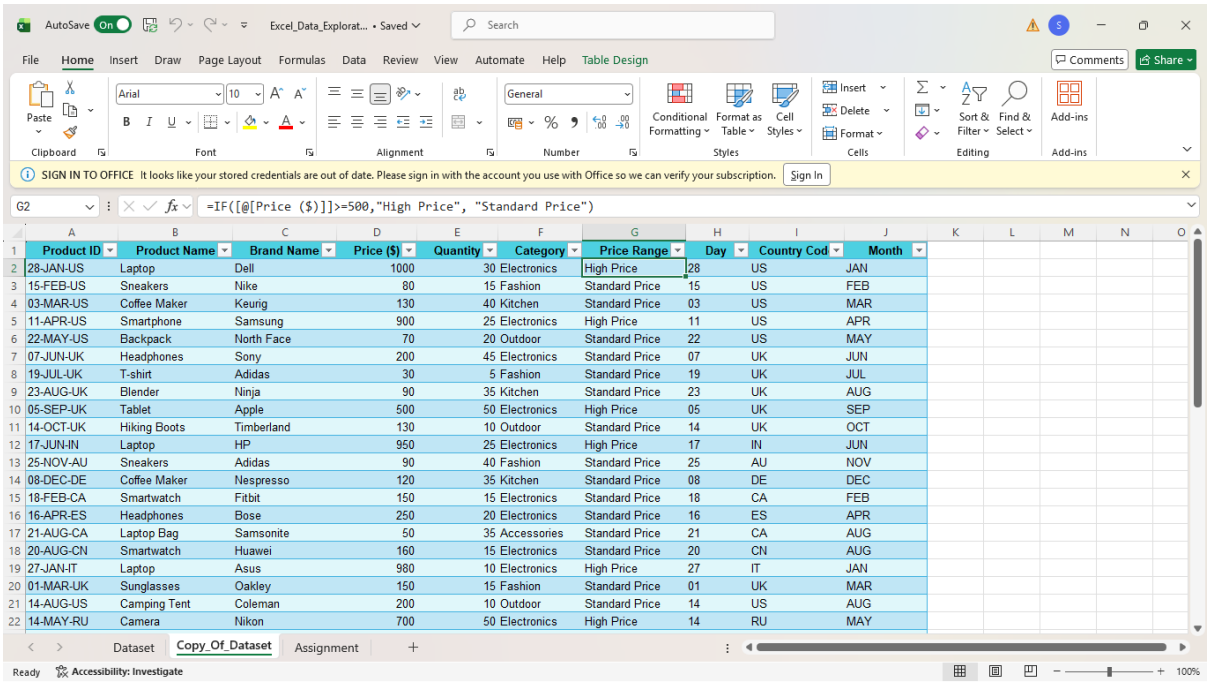
- Identify the **maximum price** among all products.

Excel Assignment 1 – Data Exploration



3. Logical Function – IF

- Create a new column named **Price Range**.
- Use the **IF function** to categorize products based on their price:
  - If **Price ≥ $500** → classify as **High Price**
  - If **Price < $500** → classify as **Standard Price**



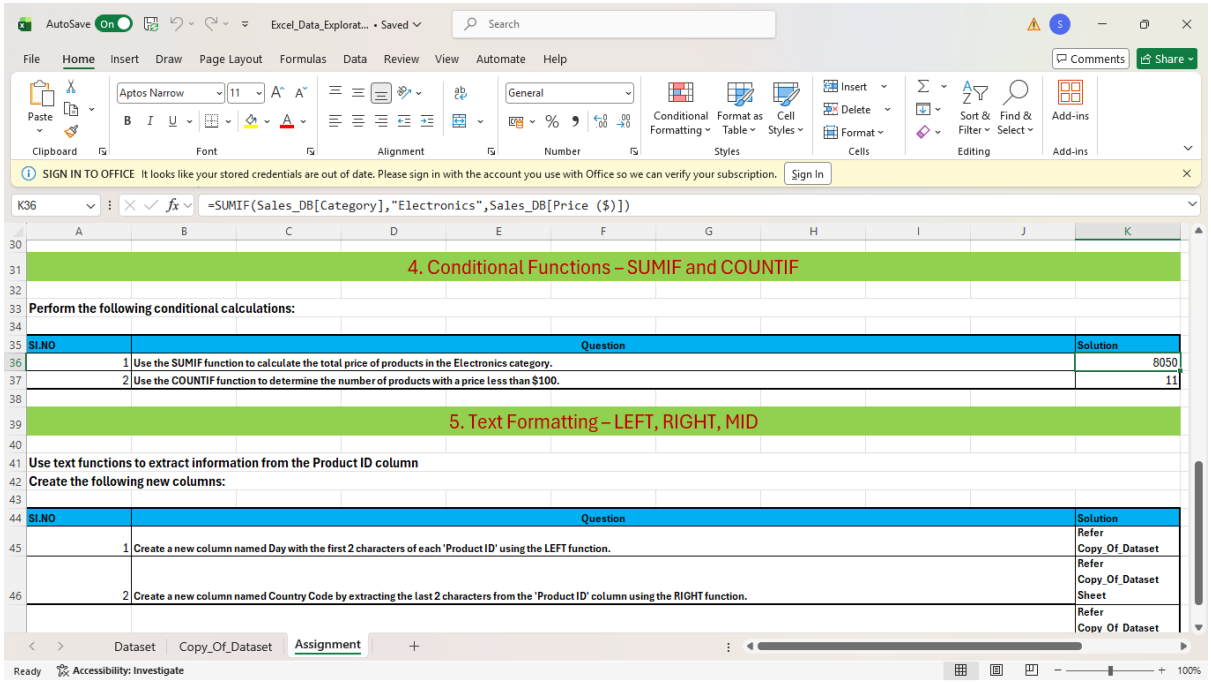
Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Day	Country Cod	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY

Excel Assignment 1 – Data Exploration

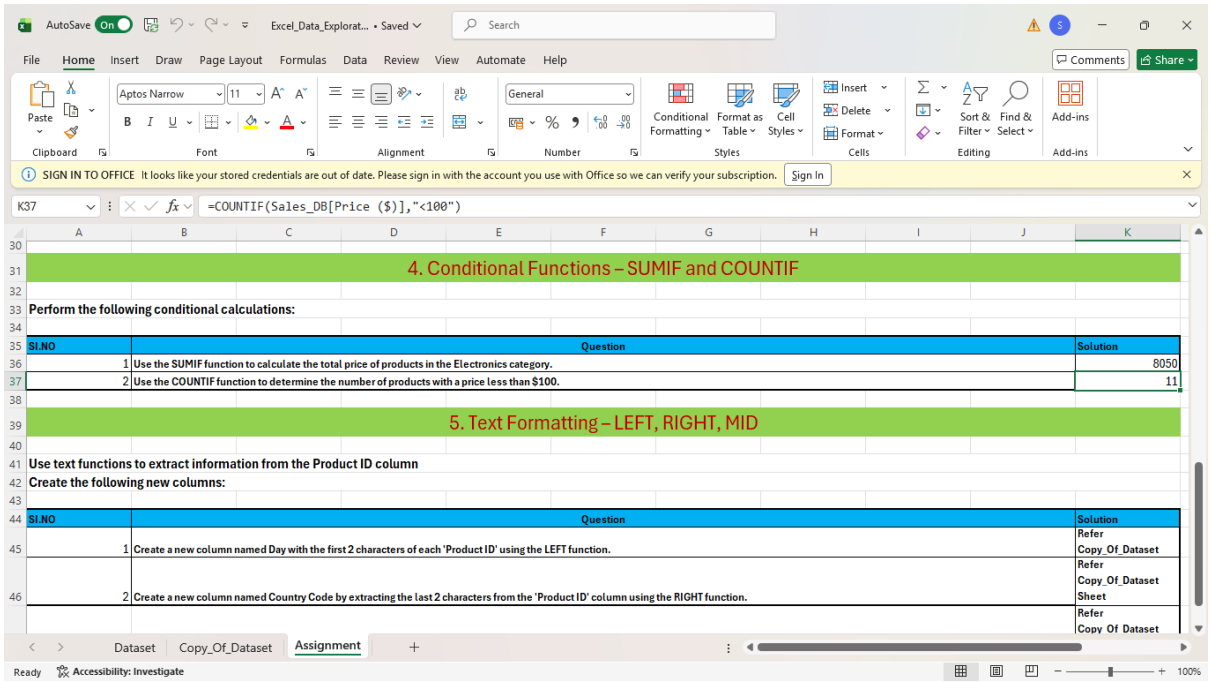
4. Conditional Functions – SUMIF and COUNTIF

Perform the following conditional calculations:

- Use the **SUMIF function** to calculate the **total price of products in the Electronics category.**



- Use the **COUNTIF function** to determine the **number of products with a price less than $100.**



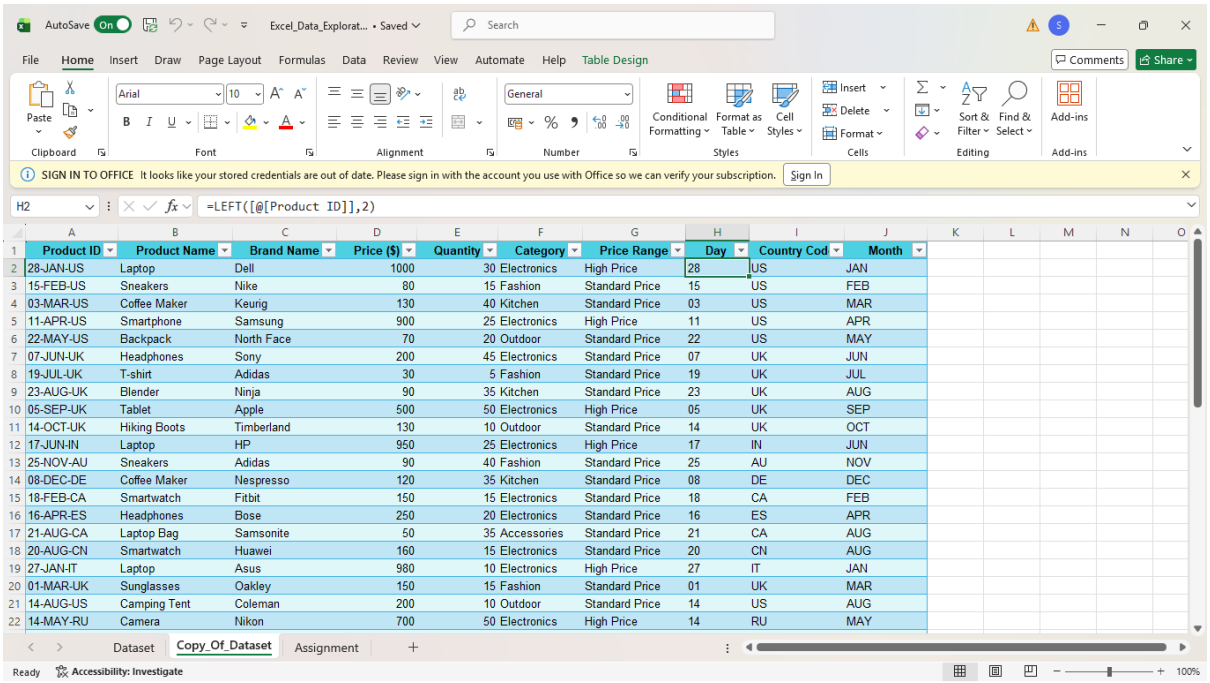
Excel Assignment 1 – Data Exploration

5. Text Formatting – LEFT, RIGHT, MID

- Use text functions to extract information from the **Product ID** column.

  Create the following new columns:

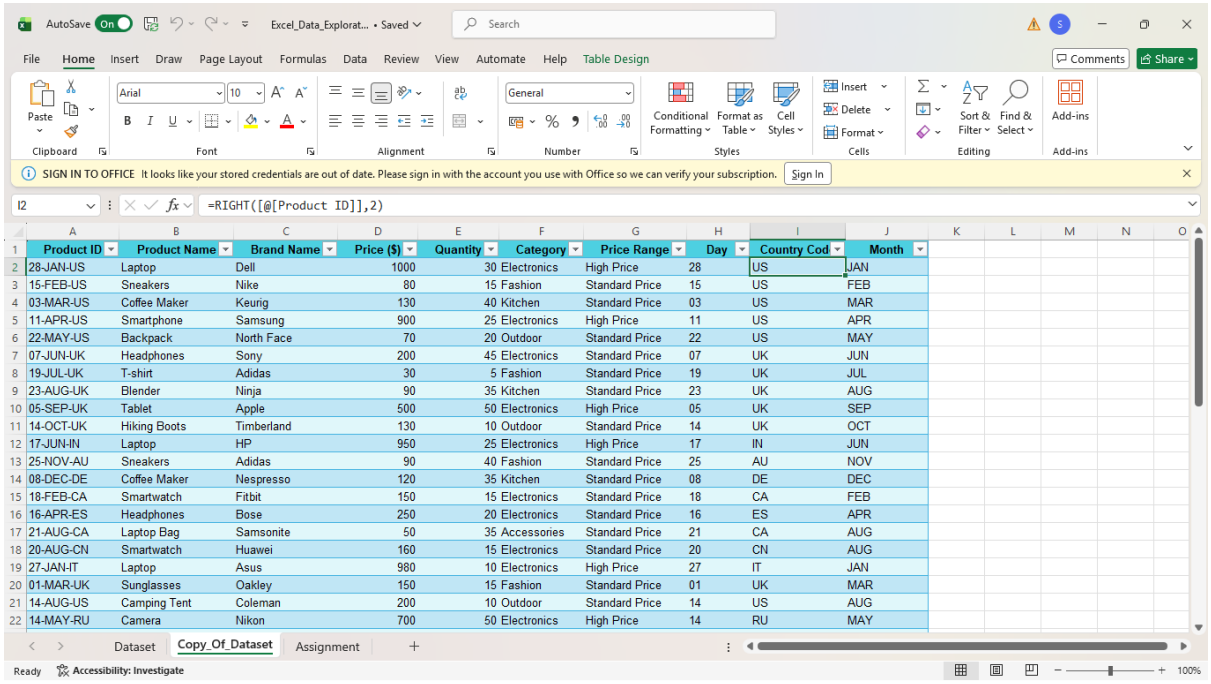
  - Create a new column named **Day** with the first 2 characters of each **'Product ID'** using the **LEFT function**.



Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Day	Country Cod	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY

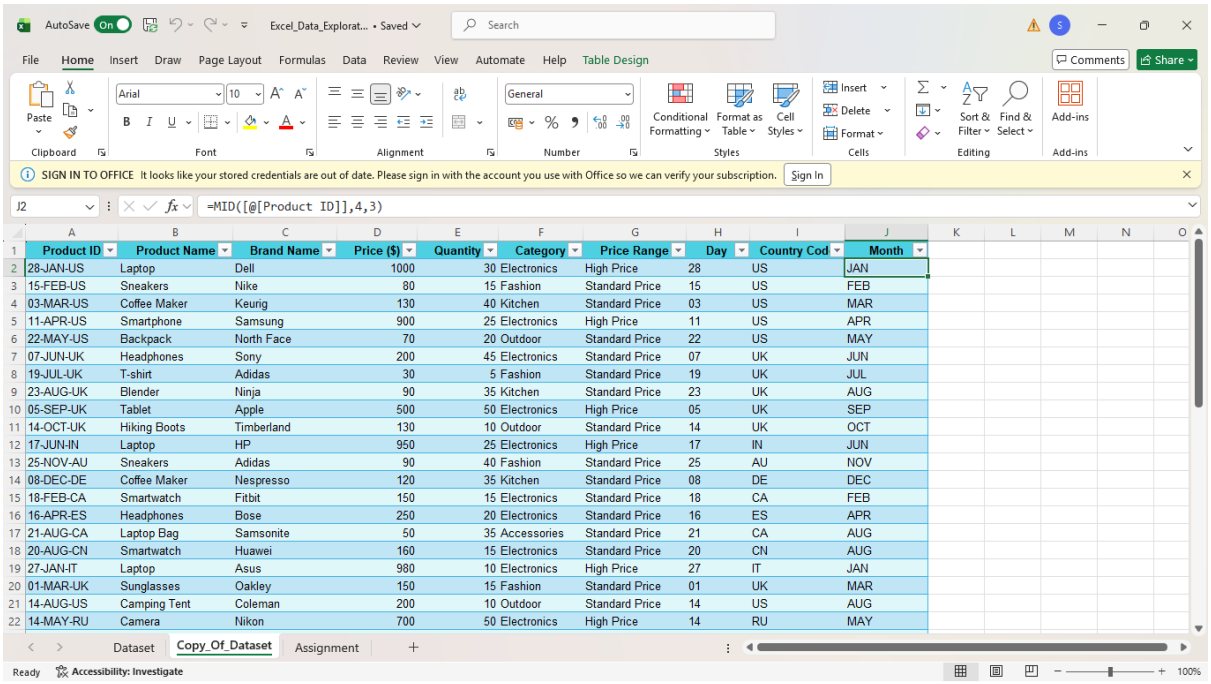
  - Create a new column named **Country Code** by extracting the last 2 characters from the **'Product ID'** column using the **RIGHT function**.

Excel Assignment 1 – Data Exploration



Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Day	Country Cod	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY

- Create a new column named **Month** by extracting the 4th to 6th characters from the **'Product ID'** column using the **MID function**.



Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range	Day	Country Cod	Month
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY